2019 EXAMINATIONS

Part I (First Year)

SCHOOL OF COMPUTING AND COMMUNICATIONS

SCC.130 Information Systems (1 hour & 30 Minutes)

- **Answer any <u>THREE</u> out of the four questions.**
- **Use a <u>separate</u> answer book for <u>each</u> question.**

QUESTION 1 (25 marks)

Consider an information system that relies on electronic data and protocol based inter-network communications as a core part of its business operation:

1.a Define what a protocol is in this context and explain why protocols are needed for such a system.

[3 marks]

1.b Suppose two users need to exchange data between two ends of the system. Using a relevant example, explain the principles of protocol layering for this scenario.

[6 marks]

1.c If the same two users are based at two endpoints A and B respectively as shown in Figure 1, assuming the direction of data transfer is from User A to User B and the system uses the TCP/IP protocol suite, show the layers used and flow of data at each of the following points: Host A, Host B, Link 1, Link 2 and Router.

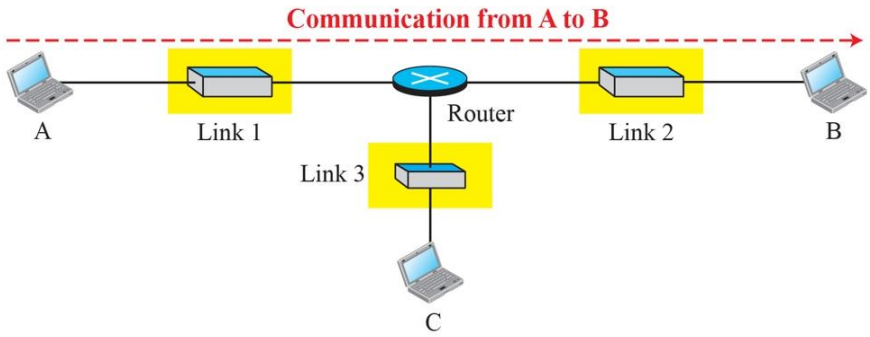

Figure 1: Two-user communication in a network based information system

[6 marks]

1.d List and compare the parameters that influence the transmission speed of data and performance of the system shown in Figure 1.

[6 marks]

Question 1 continues on next page…

Question 1 continued.

1.e Suppose the user at Host A sends data at a rate = 500 kpbs:

i. How long does it take to transmit 10 bits? **[1 mark]**

ii. What is the transmission time if Host A is sending 5 Kilo-Bytes of data? **[1 mark]**

iii. If the connection between Link 2 and Host B is 200 kbps, what is the throughput of the whole path? **[2 marks]**

[Total 25 Marks]

QUESTION 2 (25 Marks)

In this question, you will have to consider the following three relations (i.e., Airlines, Planes, and Fleets) in your answers, when applicable.

Airlines		
aid	**name**	**founded**
1	Swiss	2002-03-31
2	United	1926-04-06
3	Lufthansa	1953-01-06
4	Austrian	1957-09-30
5	Aegean	1999-05-28

Planes		
pid	**type**	**seats**
1	Airbus A340-300	291
2	Airbus A330-300	236
3	Boeing 777-300ER	340
4	Bombardier CS100	125
5	ATR 72-600	70

Fleets		
aid	**pid**	**number**
1	2	3
1	3	2
2	2	7
3	3	4
4	2	10
4	4	2
5	5	4

2.a What is a primary key? Which is the primary key in each of the above relations (i.e., Airlines, Planes, Fleets)?

[4 marks]

Question 2 continues on next page…

Question 2 continued.

2.b What are the main types of join relations? What would be the result of applying each of the join relations on two relations A and B?

[4 marks]

2.c What are the Data Definition Language (DDL), Data Manipulation Language (DML), and Data Control Language (DCL) in a Database Management System (DBMS)? Provide brief information for each including a simple example per language.

[3 marks]

2.d Write SQL statements that will generate answers to the following, including the result of them, and considering the above relations (i.e., Airlines, Planes, and Fleets).

i. Which airlines have in their fleet planes with the capacity to transport more than 250 passengers?

[3 marks]

ii. Which airline has the largest fleet (include the fleet's number of planes too)?

[5 marks]

iii. Which type(s) of planes does/do not exist in any of the fleets?

[4 marks]

iv. Show the content of the Fleets relation after executing the following SQL statement: DELETE FROM Fleets WHERE aid = 4 OR pid =2;

[2 marks]

[Total 25 Marks]

QUESTION 3 (25 marks)

3.a Define censorship, give one example of it occurring and list two risks of the Censorship.

[4 marks]

3.b The main purpose of the Data Protection Act is to control how your personal information is used by organisations, businesses or the government. State three principles of the Data Protection Act and for each principle explain how the principle affects how your personal data can be used and shared.

[6 marks]

3.c How does a Worm differ from a Trojan? Give a famous example of each and explain how they are often spread.

[5 marks]

3.d Digital Exclusion is a lack of access to, and use of, ICT resources due to different reasons. In different countries of the world, there are some people at risk of being excluded from access and use of digital technologies. State and explain three effects of digital exclusion.

[6 marks]

3.e In which contexts could a software developer's negligence have a significant effect? What steps could he/she take in order to ensure that he is not liable for negligence?

[4 marks]

[Total 25 Marks]

QUESTION 4 (25 marks)

4.a Consider the five-component framework of an information system:

i. Draw a representation of this framework, label and describe the functionality of its components.

[5 marks]

ii. Using an information system as an example, order the components in terms of increasing degree of difficulty of change and amount of disruption. Give reasons for your choice of order and how a change can affect the overall system.

[3 marks]

iii. Identify two systems where the first creates more work for people and the second creates less work. Explain how the two systems differ and the implications this has on an organisation.

[4 marks]

4.b Consider an example of a Web based information system:

i. Draw a diagram representing the layers used in the Web architecture and identify the main role played by each layer.

[6 marks]

ii. Using the same example, identify specific functions that each layer needs to provide at the development and implementation phase.

[7 marks]

[Total 25 Marks]

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